Raritone AI Demand & Financial Intelligence Platform

DEPLOYMENT DETAILS

A decentralized network of three standalone machine learning microservices designed for retail operational monitoring, macro demand forecasting, and financial trend tracking using Facebook Prophet and FastAPI.

1. Repository Blueprint & Dataset Structure

Before executing any commands, ensure your workspace folder matches this exact structural layout. All raw relational database assets must be placed inside a localized folder named `datasets`.

```
Member_2/
 ├── datasets/
 |    ├── order.csv            # Core transaction histories
 |    ├── product.csv          # Stock units and base pricing lists
 |   └── category.csv          # Relational mapping catalog
 ├── inventoryForecast.py      # App 1: Daily Units Operations (Port 8000)
 ├── productDemand.py          # App 2: 6-Month Units Projection (Port 8001)
 ├── trendForecasting.py       # App 3: 6-Month Financial Trajectory(Port 8002)
└── requirements.txt           # Consolidated system dependencies
```

2. Environment Initialization & Installation

Step 1: Clone and Navigate to the Repository

Open your terminal or PowerShell window and enter the project folder:

PowerShell:

cd Member_2

Step 2: Install System Dependencies

Execute the pip pipeline command to install the required Python frameworks, analytical math toolkits, and machine learning structures:

PowerShell

pip install -r requirements.txt

Tip for a Clean Installation: If your environment throws any compilation errors while downloading machine learning components, clear your local pip cache and force a clean installation:

pip install --upgrade --force-reinstall -r requirements.txt

3. Local Execution Guide (Running Simultaneously)

Because each microservice processes data across independent operational intervals and targets, they are configured on isolated network ports. To run all three concurrently, open three separate terminal windows side-by-side and execute the following targets:

Terminal 1: Daily Operations Forecasting Engine

Script Target: inventoryForecast.py

Port Allocation: 8000

Command:

PowerShell

python inventoryForecast.py

Interactive UI Panel: http://127.0.0.1:8000/docs

Terminal 2: 6-Month Product Demand Strategic Engine

Script Target: productDemand.py

Port Allocation: 8001

Command:

PowerShell

python productDemand.py

Interactive UI Panel: http://127.0.0.1:8001/docs

Terminal 3: 6-Month Financial & Revenue Trajectory Engine

Script Target: trendForecasting.py

Port Allocation: 8002

Command:

PowerShell

python trendForecasting.py

Interactive UI Panel: 👉 http://127.0.0.1:8002/docs

Deployment Troubleshooting Checklist

"This site can't be reached / Connection Refused"

Root Cause: The background Python backend daemon crashed or wasn't kicked off in that specific terminal frame.

Resolution: Return to the corresponding terminal window and verify that it actively displays INFO: Uvicorn running on http://127.0.0.1:XXXX.

"404 Not Found: Category has insufficient data timeline history"

Root Cause: The inputted fashion category text parameter doesn't contain at least 2 distinct transaction dates within your database file.

Resolution: Ensure database entries are populated and match standard fashion types (e.g., Kurtis, Jeans, Hoodies).

Understanding Schema "Zeroes" in the Documentation

System Behavior: The documentation panel displays a static mock template showing default 0 and "string" parameters under structural declarations. This is expected. True calculated results will compute dynamically upon clicking "Try it out" -> "Execute".